

Artificial Intelligence

SITAMS CSE (AI & ML) • R23 • II Year • III Semester • Course Code: 23CAI231T

Source basis: Official SITAMS R23 CSE (AI & ML) syllabus uploaded for this Study Hub. This concise revision guide preserves the syllabus topic statements and adds a study checklist; it is not a replacement for prescribed textbooks or faculty notes.

Unit I: INTRODUCTION

Syllabus topics

- AI problems, foundation of AI and history of AI intelligent agents: Agents and Environments, the concept of rationality, the nature of environments, structure of agents, problem solving agents, problem formulation

Revision checklist

- Know the definitions and principles named in the syllabus.
- Prepare the standard methods, diagrams, formulas or algorithms relevant to the listed topics.
- Practice representative examples or problems from the prescribed study material.
- Review this unit against the official SITAMS syllabus before attempting its quiz.

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Unit 2: SEARCHING

Syllabus topics

- Searching for solutions, uniformed search strategies – Breadth first search, depth first Search
- Search with partial information (Heuristic search) Hill climbing, A* ,AO* Algorithms, Problem reduction, Game Playing-Adversial search, Games, mini-max algorithm, optimal decisions in multiplayer games, Problem in Game playing, Alpha-Beta pruning, Evaluation functions

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Unit 3: REPRESENTATION OF KNOWLEDGE

Syllabus topics

- Knowledge representation issues, predicate logic- logic programming, semantic nets- frames and inheritance, constraint propagation, representing knowledge using rules, rules based deduction systems
- Reasoning under uncertainty, review of probability, Bayes■ probabilistic interferences and dempstershafer theory

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Unit 4: LOGIC CONCEPTS

Syllabus topics

- First order logic
- Inference in first order logic, propositional vs
- first order inference, unification & lifts forward chaining, Backward chaining, Resolution, Learning from observation Inductive learning, Decision trees, Explanation based learning, Statistical Learning methods, Reinforcement Learning

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Unit 5: EXPERT SYSTEMS

Syllabus topics

- Architecture of expert systems, Roles of expert systems – Knowledge Acquisition Meta knowledge Heuristics
- Typical expert systems – MYCIN, DART, XCON: Expert systems shells

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Study Hub status

This guide is syllabus-derived. External resources and quiz questions are tracked separately and are only labelled verified after source checking.